arXiv:1703.04933v2 [cs.LG] 15 May 2017

Sharp Minima Can Generalize For Deep Nets

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Abstract

Despite their overwhelming capacity to overfit, deep learning architectures tend to generalize relatively well to unseen data, allowing them to be deployed in practice. However, explaining why this is the case is still an open area of research. One standing hypothesis that is gaining popularity, e.g. Hochreiter & Schmidhuber (1997); Keskar et al. (2017), is that the flatness of minima of the loss function found by stochastic gradient based methods results in good generalization. This paper argues that most notions of flatness are problematic for deep models and can not be directly applied to explain generalization. Specifically, when focusing on deep networks with rectifier units, we can exploit the particular geometry of parameter space induced by the inherent symmetries that these architectures exhibit to build equivalent models corresponding to arbitrarily sharper minima. Furthermore, if we allow to reparametrize a function, the geometry of its parameters can change drastically without affecting its generalization properties.

1 Introduction

Deep learning techniques have been very successful in several domains, like *object recognition in images* (e.g Krizhevsky et al., 2012; Simonyan & Zisserman, 2015; Szegedy et al., 2015; He et al., 2016), *machine translation* (e.g. Cho et al., 2014; Sutskever et al., 2014; Bahdanau et al., 2015; Wu et al., 2016; Gehring et al., 2016) and *speech recognition* (e.g. Graves et al., 2013; Hannun et al., 2014; Chorowski et al., 2015; Chan et al., 2016; Collobert et al., 2016). Several arguments have been brought forward to justify these empirical results. From a representational point of view, it has been argued that deep networks can efficiently approximate certain functions (e.g. Montufar et al., 2014; Raghu et al., 2016). Other works (e.g Dauphin et al., 2014; Choromanska et al., 2015) have looked at the structure of the error surface to analyze how trainable these models are. Finally, another point of discussion is how well these models can generalize (Nesterov & Vial, 2008; Keskar et al., 2017; Zhang et al., 2017). These correspond, respectively, to low *approximation*, *optimization* and *estimation* error as described by Bottou (2010).

Our work focuses on the analysis of the estimation error. In particular, different approaches had been used to look at the question of why *stochastic gradient descent* results in solutions that generalize well (Bottou & LeCun, 2005; Bottou & Bousquet, 2008). For example, Duchi et al. (2011); Nesterov & Vial (2008); Hardt et al. (2016); Bottou et al. (2016); Gonen & Shalev-Shwartz (2017) rely on the concept of *stochastic approximation* or *uniform stability* (Bousquet & Elisseeff, 2002). Another conjecture that was recently (Keskar et al., 2017) explored, but that could be traced back to Hochreiter & Schmidhuber (1997), relies on the geometry of the loss function around a given solution. It argues that flat minima, for some definition of flatness, lead to better generalization. Our work focuses on this particular conjecture, arguing that there are critical issues when applying the concept of flat minima to deep neural networks, which require rethinking what flatness actually means.

While the concept of flat minima is not well defined, having slightly different meanings in different works, the intuition is relatively simple. If one imagines the error as a one-dimensional curve, a minimum is flat if there is a wide region around it with roughly the same error, otherwise the minimum is sharp. When moving to higher dimensional spaces, defining flatness becomes more complicated. In Hochreiter & Schmidhuber (1997) it is defined as the size of the connected region around the minimum where the training loss is relatively similar. Chaudhari et al. (2017) relies, in contrast, on the curvature of the second order structure around the minimum, while Keskar et al. (2017) looks at the maximum loss in a bounded neighbourhood of the minimum. All these works rely on the fact that flatness results in robustness to low precision arithmetic or noise in the parameter space, which, using an *minimum description length*-based argument, suggests a low expected overfitting.

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Proceedings of the 34th International Conference on Machine Learning, Sydney, Australia, 2017. JMLR: W&CP. Copyright 2017 by the author(s).

However, several common architectures and parametrizations in deep learning are already at odds with this conjecture, requiring at least some degree of refinement in the statements made. In particular, we show how the geometry of the associated parameter space can alter the ranking between prediction functions when considering several measures of *flatness/sharpness*. We believe the reason for this contradiction stems from the Bayesian arguments about KL-divergence made to justify the generalization ability of flat minima (Hinton & Van Camp, 1993). Indeed, *Kullback-Liebler divergence is invariant to change of parameters whereas the notion of "flatness" is not*. The demonstrations of Hochreiter & Schmidhuber (1997) are approximately based on a Gibbs formalism and rely on strong assumptions and approximations that can compromise the applicability of the argument, including the assumption of a discrete function space.

2 Definitions of flatness/sharpness

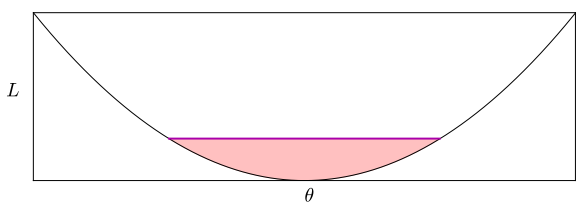

Figure 1: An illustration of the notion of flatness. The loss L as a function of θ is plotted in black. If the height of the red area is ϵ, the width will represent the volume ϵ-flatness. If the width is 2ϵ, the height will then represent the ϵ-sharpness. Best seen with colors.

For conciseness, we will restrict ourselves to supervised scalar output problems, but several conclusions in this paper can apply to other problems as well. We will consider a function f that takes as input an element x from an input space $\mathcal{X}$ and outputs a scalar y. We will denote by f_θ the prediction function. This prediction function will be parametrized by a parameter vector θ in a parameter space Θ. Often, this prediction function will be over-parametrized and two parameters $(\theta, \theta') \in \Theta^2$ that yield the same prediction function everywhere, $\forall x \in \mathcal{X}, f_\theta(x) = f_{\theta'}(x)$, are called *observationally equivalent*. The model is trained to minimize a continuous loss function L which takes as argument the prediction function f_θ. We will often think of the loss L as a function of θ and adopt the notation $L(\theta)$.

The notion of flatness/sharpness of a minimum is relative, therefore we will discuss metrics that can be used to compare the relative flatness between two minima. In this section we will formalize three used definitions of flatness in the literature.

Hochreiter & Schmidhuber (1997) defines a flat minimum as "a large connected region in weight space where the error remains approximately constant". We interpret this formulation as follows:

Definition 1. *Given $\epsilon > 0$, a minimum θ, and a loss L, we define $C(L, \theta, \epsilon)$ as the largest (using inclusion as the partial order over the subsets of Θ) connected set containing θ such that $\forall \theta' \in C(L, \theta, \epsilon), L(\theta') < L(\theta) + \epsilon$. The ϵ-flatness will be defined as the volume of $C(L, \theta, \epsilon)$. We will call this measure the volume ϵ-flatness.*

In Figure 1, $C(L, \theta, \epsilon)$ will be the purple line at the top of the red area if the height is ϵ and its volume will simply be the length of the purple line.

Flatness can also be defined using the local curvature of the loss function around the minimum if it is a critical point [1]. Chaudhari et al. (2017); Keskar et al. (2017) suggest that this information is encoded in the eigenvalues of the Hessian. However, in order to compare how flat one minimum versus another, the eigenvalues need to be reduced to a single number. Here we consider the *spectral norm and trace of the Hessian*, two typical measurements of the eigenvalues of a matrix.

Additionally Keskar et al. (2017) defines the notion of ϵ-sharpness. In order to make proofs more readable, we will slightly modify their definition. However, because of norm equivalence in finite dimensional space, our results will transfer to the original definition in full space as well. Our modified definition is the following:

Definition 2. *Let $B_2(\epsilon, \theta)$ be an Euclidean ball centered on a minimum θ with radius ϵ. Then, for a non-negative valued loss function L, the ϵ-sharpness will be defined as proportional to*

$$\frac{\max_{\theta' \in B_2(\epsilon,\theta)} \big(L(\theta') - L(\theta)\big)}{1 + L(\theta)}.$$

In Figure 1, if the width of the red area is 2ϵ then the height of the red area is $\max_{\theta' \in B_2(\epsilon,\theta)} \big(L(\theta') - L(\theta)\big)$.

ϵ-sharpness can be related to the spectral norm of the Hessian. Indeed, a second-order Taylor expansion of L around a critical point minimum is written

$$L(\theta') = L(\theta) + \frac{1}{2} (\theta' - \theta) (\nabla^2 L)(\theta)(\theta' - \theta)^T + o(\|\theta' - \theta\|_2^2).$$

In this second order approximation, the ϵ-sharpness at θ

[1] In this paper, we will often assume that is the case when dealing with Hessian-based measures in order to have them well-defined.

would be

$$\frac{\left|\left|\left|(\nabla^2 L)(\theta)\right|\right|\right|_2 \epsilon^2}{2\big(1+L(\theta)\big)}.$$

3 Properties of Deep Rectified Networks

Before moving forward to our results, in this section we first introduce the notation used in the rest of paper. Most of our results, for clarity, will be on the deep rectified feedforward networks with a linear output layer that we describe below, though they can easily be extended to other architectures (e.g. convolutional, etc.).

Definition 3. *Given K weight matrices $(\theta_k)_{k\leq K}$ with $n_k = dim\big(vec(\theta_k)\big)$ and $n = \sum_{k=1}^{K} n_k$, the output y of a deep rectified feedforward networks with a linear output layer is:*

$$y = \phi_{rect}\Big(\phi_{rect}\big(\cdots\phi_{rect}(x\cdot\theta_1)\cdots\big)\cdot\theta_{K-1}\Big)\cdot\theta_K,$$

where

- *x is the input to the model, a high-dimensional vector*
- *ϕ_{rect} is the rectified elementwise activation function (Jarrett et al., 2009; Nair & Hinton, 2010; Glorot et al., 2011), which is the positive part $(z_i)_i \mapsto (\max(z_i, 0))_i$.*
- *vec reshapes a matrix into a vector.*

Note that in our definition we excluded the bias terms, usually found in any neural architecture. This is done mainly for convenience, to simplify the rendition of our arguments. However, the arguments can be extended to the case that includes biases (see Appendix B). Another choice is that of the linear output layer. Having an output activation function does not affect our argument either: since the loss is a function of the output activation, it can be rephrased as a function of linear pre-activation.

Deep rectifier models have certain properties that allows us in section 4 to arbitrary manipulate the flatness of a minimum.

An important topic for optimization of neural networks is understanding the non-Euclidean geometry of the parameter space as imposed by the neural architecture (see, for example Amari, 1998). In principle, when we take a step in parameter space what we expect to control is the change in the behavior of the model (i.e. the mapping of the input x to the output y). In principle we are not interested in the parameters per se, but rather only in the mapping they represent.

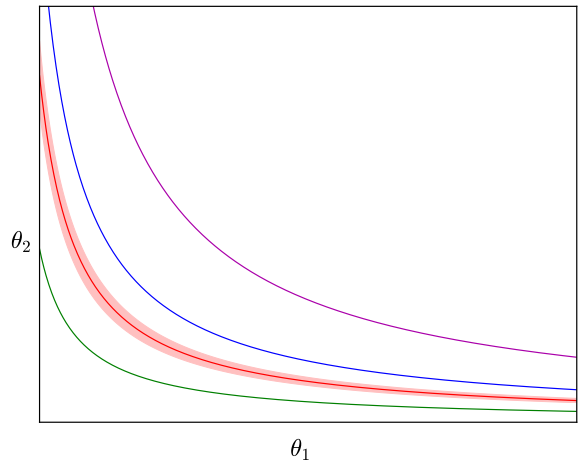

Figure 2: An illustration of the effects of non-negative homogeneity. The graph depicts level curves of the behavior of the loss L embedded into the two dimensional parameter space with the axis given by θ_1 and θ_2. Specifically, each line of a given color corresponds to the parameter assignments (θ_1, θ_2) that result observationally in the same prediction function f_θ. Best seen with colors.

If one defines a measure for the change in the behavior of the model, which can be done under some assumptions, then, it can be used to define, at any point in the parameter space, a metric that says what is the equivalent change in the parameters for a unit of change in the behavior of the model. As it turns out, for neural networks, this metric is not constant over Θ. Intuitively, the metric is related to the curvature, and since neural networks can be highly non-linear, the curvature will not be constant. See Amari (1998); Pascanu & Bengio (2014) for more details. Coming back to the concept of flatness or sharpness of a minimum, this metric should define the flatness.

However, the geometry of the parameter space is more complicated. Regardless of the measure chosen to compare two instantiations of a neural network, because of the structure of the model, it also exhibits a large number of symmetric configurations that result in exactly the same behavior. Because the rectifier activation has the non-negative homogeneity property, as we will see shortly, one can construct a continuum of points that lead to the same behavior, hence the metric is singular. Which means that one can exploit these directions in which the model stays unchanged to shape the neighbourhood around a minimum in such a way that, by most definitions of flatness, this property can be controlled. See Figure 2 for a visual depiction, where the flatness (given here as the distance between the different level curves) can be changed by moving along the curve.